

STATEMENT OF WORK (SOW)

ECAO Chiller Assembly Replacement – Full & Open Competition

Supply with Incidental Installation

1. Overview

The Bureau of Reclamation, Eastern Colorado Area Office (ECAO), requires the purchase and installation of a new water-cooled chiller, equivalent in function and performance to the existing 50-ton unit currently serving the facility. The chiller to be replaced is a 50-ton raw water-cooled reciprocating liquid chiller Carrier Model 30HK-050-D-610 Serial 2801Q00606. The heat removal from the chiller is accomplished by piped flowing water which is then discharged to a lake. Water from the same source is cooled by the chiller refrigeration and is piped to the office work spaces for temperature conditioning.

The ECAO main office is located at 11056 West County Road 18E, Loveland CO 80537, approximately 5-miles west of Loveland CO.

This procurement will be completed as a supply buy with incidental installation.

2. Scope of Work

The Contractor shall furnish all labor, supervision, equipment, tools, transportation, and materials necessary to supply and install a new water-cooled chiller, remove the existing chiller including R-22 disposal, complete mechanical/electrical tie-ins, and provide startup/commissioning. As part of the scope of work the contractor shall perform a pre purchase site visit and report. The contractor shall visit the site prior to purchasing the replacement chiller to field verify all pertinent installation information including, but not limited to power wiring, power feed, control wiring, control interfaces and control points, supply water piping and flow rates, water-to-work space temperature conditioning units piping and flow rates, and available space for equipment placement. Contractor shall verify the suitability of the proposed replacement chiller for the application. Contractor shall report on these findings in submittal in **Required Submittal Number (RSN) 14**.

3. Salient Characteristics

- **General:** Chiller, water cooled, nominal 50-ton rating, designed for commercial building HVAC applications with a 100% duty cycle rating during summer daylight hours to be located in an indoor chiller mechanical room
- **Electrical Requirement:** 460 VAC, 3 phase, 60 Hz; The existing unit is designed for minimum supply circuit of 75-amps, with a maximum overcurrent protection of 100-amps and is currently fed from a 100-amp breaker. The replacement chiller current draw may be less than that of the existing chiller, but not more.

- **Refrigerant Requirement:** American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) Standard 34, Designation and Safety Classification of Refrigerants, Classification A1 is preferred. A2L refrigerant may be considered after submittal review provided that a refrigerant detection system compliant with ASHRAE 34 and UL 60335-2-40 standards is provided as part of the lump sum cost. Refrigerant shall be legal for manufacture in the United States (40 Code of Federal Regulations, CFR, Title 40, Chapter 1, Subchapter C, Part 84, Subpart B 84.54 restrictions on the use of hydrofluorocarbons).
- **Noise Requirement** – 70dB or less inside the chiller mechanical room when the chiller is operating, but the air handler is not running.
- **Control, Alarm, and Diagnostic Panel:** The chiller shall be equipped with the manufacturer's built-in control, diagnostic, and alarm panel for intuitive operation of the chiller as integrated equipment as it comes standard from the manufacturer. An "off the shelf" unit is desired so some variance between manufacturers is allowed. The capabilities of the existing chiller panel are included in Appendix XXX.
- **Energy Efficiency Requirements:** Equipment shall comply with Federal Energy Management Program (FEMP) (42 U.S.C. 8259b) (10 CFR part 436, subpart C) and ASHRAE Standard 90.1 (most recent).
- **Exchanger Cleaning:** Water side of any heat exchange equipment shall be accessible for routine maintenance cleaning without disassembly or movement of the chiller. Contractor shall explain these cleaning procedures during staff training, below.
- **Chiller Installation Pad Space Limitations:** The contractor shall install the new chiller assembly (chiller) into the same space that the existing chiller is to be removed from. This space measures 9-feet long by 5-feet wide by 6-feet, 6-inches high. It sits on a concrete equipment pad that is 5-feet long by 4-feet wide by 3-inches high. Additional space is available on the long dimension if a full 5-feet wide rectangular shape is not required. While the height of the ceiling in the chiller room is 10-feet, 3-inches, the height of the available space is impacted by piping and ductwork. The existing chiller sits with the long dimension running north-south in the room. The door to the room is adjacent to the southwest corner of the chiller.
- **Mounting:** Chiller shall be mounted by permanent concrete anchors to the concrete equipment pad. The equipment pad shall be increased in size as necessary by the contractor to accommodate the replacement chiller mounting. The steel floor grating surrounding the equipment pad shall be modified as necessary by the contractor to maintain the original outside shape and function. Contractor modifications to the connecting piping, wiring, concrete pad, drains, and miscellaneous equipment shall be included in the lump sum price.
- **Vibration Dampening / Isolation:** The contractor shall install appropriate vibration dampening so that the current condition of no perceptible vibration being transmitted to the office building structure is maintained. The chiller rests on coiled spring vibration dampeners, which the contractor shall functionally replicate on the

new chiller. Refer to Appendix 2 - Drawing No. 245-D-10064, and Appendix 9 Vibration Dampening & Cleanout Details attached, but field verify.

- **Manuals & Drawings:** Contractor shall provide a complete set of equipment drawings, operating and maintenance manuals, and warranty information from the manufacturer. **RSN 17**
- **Training:** Contractor shall provide in person training to Reclamation mechanical staff sufficient for operation and maintenance. Documented in **RSN 17**
- **Warranty:** Contractor will provide a minimum of a one-year warranty on all work, parts, and labor in addition to establishing for Reclamation any standard manufacturer's warranties.

4. Removal and Disposal of Existing Unit

The existing Carrier Model 30HK-050-D-610 Serial 2801Q00606 chiller assembly and associated piping, fittings, refrigerant, and ancillary parts shall be removed from government property and properly disposed of, recycled, or salvaged by the contractor. The equipment and refrigerant shall be handled pursuant to EPA regulations (40 CFR Part 82, Subpart F) under Section 608 of the Clean Air Act which include requirements for the safe disposal of refrigeration and air-conditioning equipment. The smallest door through which the equipment must pass is a rectangular opening 4-feet, 10-inches wide by 6-feet, 8-inches high. The basement in which the chiller resides has an outside floor level access ramp that accommodates standard sized pick-up trucks. The existing chiller assembly is approximately 9-feet long, by 3-feet wide, by 6-feet high. The contractor shall provide any necessary lifting or transport devices for removal and installation. The nameplate data for the existing chiller is shown in Appendix 1. Views of the existing chiller are shown in Appendices 9 - 12.

5. Installation Requirements

The objective of this procurement is to replace an old worn chiller packaged unit with a newly manufactured "off-the-shelf" chiller of equal or minimally superior capabilities and to ensure it works as intended in the building HVAC system in the same location from which the old one is removed. Reciprocating compression is not a requirement. Quieter compression methods may be proposed by the contractor. The contract price shall also include any equipment, materials, or supplies necessary to accomplish this. If because of differences between the new chiller and the old chiller, additional pumps, filtration, or other equipment are required, these items shall be included in the contract price. The nameplate data and service manuals that detail the capabilities and specifications of the old, existing chiller are attached as Appendices 1, 7, and 8.

The contractor shall install the new chiller assembly (chiller) into the same space that the existing chiller is to be removed from. Additionally, the contractor shall wire the chiller into

the existing electrical panel that resides on the wall just north of the chiller. The contractor shall connect all entering and exiting water lines to the existing lines and shall provide any necessary pipe, pipe connections, or ancillary parts to accomplish this. Galvanically compatible materials that will prevent corrosion shall be used throughout. Removable pipe fittings shall be used to connect to building piping, except if not technically feasible, or due to space restriction. Water piping enters the space at both the north and south ends. Refer to Appendix 3 - Drawing Nos. 245-D-10066, and Appendix 4 - 245-713-2122, Appendix 5 - 245-D-10067, and Appendix 6 – Drawing 245-D-10062 attached, but field verify.

The existing chiller has performed adequately with the sourced water which is untreated, minimally filtered lake water that includes seasonally variable amounts of sediment and biological debris. If the new chiller provided requires additional filtration by the manufacturer for proper function or warranty compliance, then the contractor shall provide and install the proper filtration as part of the contract cost. The contractor shall during installation ensure that all water flows to the chiller for heat removal and to the building for temperature conditioning are within manufacturer recommended ranges. If the pressure drops across the newly installed equipment are higher than the existing, the contractor shall install pumps as necessary to ensure proper function of the building cooling system as part of the contract cost.

Additionally, the contractor shall connect the new chiller to the building HVAC control programmable logic controller (PLC) and confirm proper functioning of the interface. The existing interface consists of start-stop control wiring and an on-off alarm point. The PLC is a Solidyne Schnider Electric product.

6. Startup, Testing, Commissioning, Post Installation Service

- **Commissioning** – Contractor shall perform commissioning per manufacturer's recommendations and shall ensure proper control and alarming interfacing with the building HVAC control programmable logic controller (PLC). Additionally, contractor shall verify refrigerant charge, electrical performance, and system functionality. Contractor shall report on these findings in submittal **RSN 15**.
- **Post Installation Service** – Contractor shall perform winterization and summarization inspections and chiller adjustments each November and each May for the next two months of November and May following installation for a total of 4 inspection / adjustment visits.
- **Response Time** – Contractor shall maintain capability for weekday 4-hour response time to the building location to provide a competent technician with a

service truck stocked sufficiently to perform minor repairs or adjustments from installation to the end of all contractor or manufacturer warranties.

7. Submittals

Contractor shall submit all required submittals pursuant to the following Submittal Table. In addition, contractor shall submit the following as **RSN 16**

- Parts list of all supplied pieces of equipment, pipe, fittings, but excluding miscellaneous supplies,
- Chiller Specification sheets, including manufacturer specified dB rating & duty cycle rating,
- Description of chiller control, alarm, and diagnostic integrated panel functions and capabilities,
- Refrigerant specification sheet,
- Ancillary equipment, if any, specification sheets,
- Equipment layout drawing also including any site modifications and vibration dampening,
- Historical length of time the manufacturer has produced this size & type of chiller,
- Locations of chiller manufacturer maintenance/technical support offices closest to the installation site.

The Contractor shall furnish submittals in accordance with the following table:

** CO indicates Contracting Officer; COR indicates Contracting Officer's Representative.

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible	No. of sets to be sent to: **	
					CO	COR
1	Bonds 52.228-15	Performance and Payment Bonds	Within 10 calendar days after award	CO	1 electronic	0

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible	No. of sets to be sent to: **	
					CO	COR
2	Additional Bond Security 52.228-2	Additional Bond Security	Within 10 calendar days after award	CO	1	0
3	Insurance- work on a Government Installation 1452.228- 70	1) Written certification that the required insurance has been obtained 2) Current certification of insurance for each subcontractor	1) Before issue of notice to proceed 2) Prior to Contractors commencing onsite work	CO	1 electronic	0
4	Liability Insurance 1452.228- 70	Acceptable Certificate of Liability Insurance	Prior to commencement of work under the contract	CO	1 electronic	0
5	Equal Opportunity 52.222-26	Information required by Executive Order 112246 (SF-100)	Within 28 calendar days after award	CO	1 electronic	0

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible	No. of sets to be sent to: **	
					CO	COR
6	Reporting Executive Compensation and First Tier Subcontract Awards 52.204-10	Certification that executive compensation and subcontract awards were reported	Within 28 days of each report being submitted	CO	1 electronic	0
7	Schedule of Values GP-1	Required for lump- sum items in the Bid Schedule	Submitted and approved before any payment other than bonds and insurance	CO	1 electronic	0
8	Invoice DOI – AAAP 0028 Electronic Invoicing and Payment Requirements (IPP)	DRAFT Invoice before entering into IPP	10 Days Prior to Payment Request	CO	1 electronic	
9	Release of Claims 1452.204- 70	Release of Claims (DI-137) against the Government	After completion of the work and prior to final payment	CO	1 electronic	0

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible	No. of sets to be sent to: **	
					CO	COR
10	Site Security GP-7	List of all contractor and subcontractor employees	At least 28 days before mobilization	COR	0	1 paper or 1 encrypted electronic
11	Site Security GP-7	List of all vehicles and equipment to be brought onsite	At least 28 days before mobilization	COR	0	1 paper or 1 encrypted electronic
12	Labor Standards Data 52.222-11	1) List of subcontractors 2) Statement and Acknowledgement Form (SF-1413) for each subcontract	At least 28 days before mobilization	COR	1 electronic	1 paper or 1 encrypted electronic
13	Payrolls 52.222-08	Copies of all payrolls to CO	Weekly for each week of contract performance	CO	1 electronic encrypted	1 paper or 1 encrypted electronic

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible	No. of sets to be sent to: **	
					CO	COR
14	2.0 Scope/Site Visits/Pre- Installation	Preinstallation site visit report.	Approved prior to chiller purchase	COR	0	1 electronic
15	2.0 Scope/Site Visits/Com missioning	Commissioning Report	Approved prior to final payment	COR	0	1 electronic
16	2.0 Scope/Site Visits/Techn ical Submittals	As required by section text	Approved prior to chiller purchase			1 electronic

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible	No. of sets to be sent to: **	
					CO	COR
17	4.0 Work Requirements	<u>Final Documentation:</u> All available manufacturer documentation for Chiller, ancillary equipment, parts, and materials installed under this contract, including Operation and Maintenance (O&M) manuals. At a minimum, this shall include final wiring diagrams, maintenance data, final shop drawings available manufacturer drawings, training manuals, and warranty documents.	Within 30 days of on-site work completion.	COR	0	1 electronic
18	4.0 Work Requirements	<u>Final Report:</u> Final report detailing additional issues found during work and recommendations for future work.	Within 30 days of on-site work completion.	COR	0	1 electronic

8. Period of Performance

All on-site installation and commissioning work shall be completed within 180 days of award. Follow-up seasonal inspections and adjustment visits shall be completed pursuant to 2.0 SCOPE / Site Visits and 6.0 Startup, Testing, Commissioning, Post Installation Service, above.

9. Government Responsibilities

Provide site access, utility coordination, COR oversight. The Government will provide basic 110v power as needed for tools used during the installation at the powerplant.

10. Reference Appendices (Reference Only)

Appendices 1–12 are included for reference only and do not establish brand-specific requirements.

11. Daily Work Hours

All onsite work is required to be done during the following timeframes. Monday through Friday 6:30 AM to 3:00 PM. No work will be done on weekends, or on Federal Holidays. Reclamation personnel will be present during work. Installation must be scheduled for a minimum of 2 weeks prior to work commencing.

12. Damages

Contractor is responsible for all property damage incurred to Reclamation property during installation.